

University of Dhaka  
Affiliated Engineering Colleges  
Department of Computer Science and Engineering  
1<sup>st</sup> Year 1<sup>st</sup> Semester B.Sc. Examination, 2024  
CSE – 1102, Discrete Mathematics

Total Marks: 70

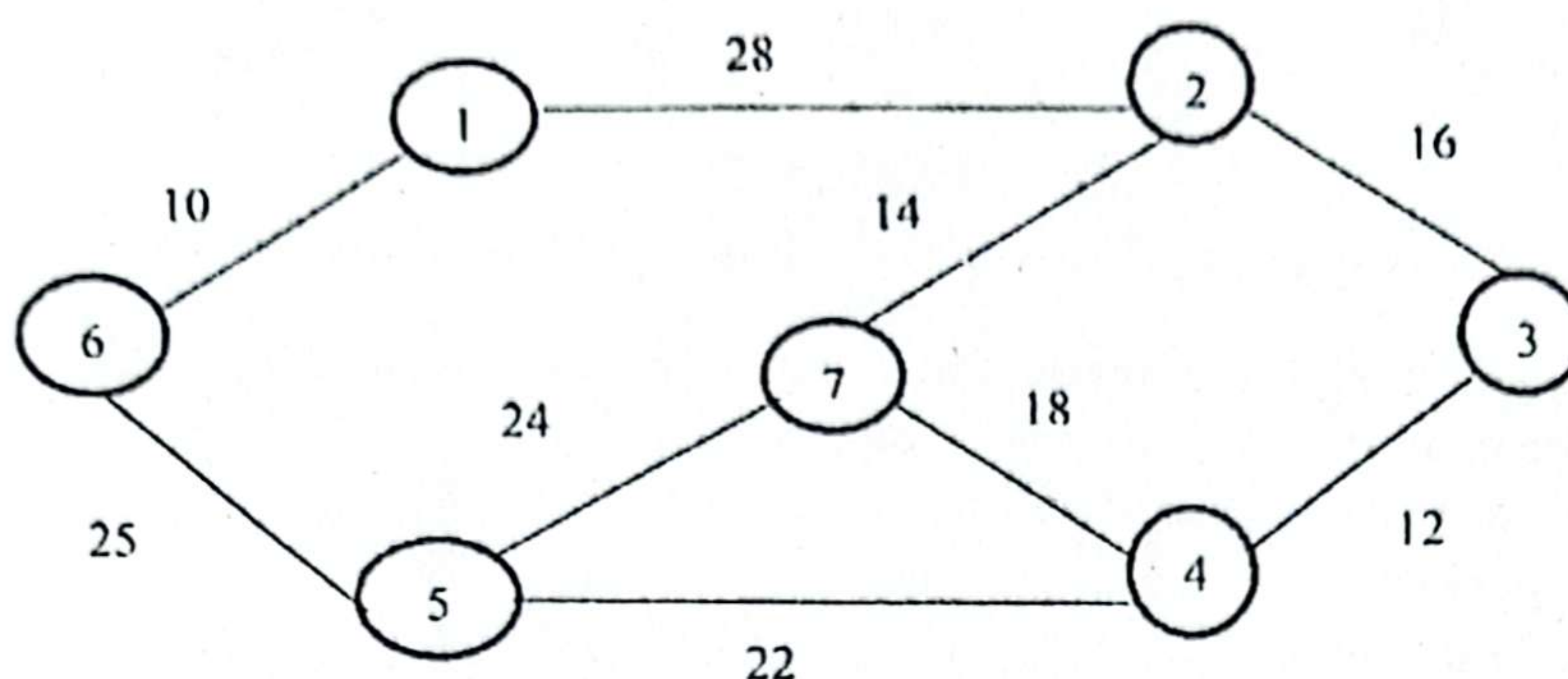
Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) What is Quantifier? Represent the following sentences using the above predicates, appropriate quantifiers and logical connectives. 1+4  
**Domain: Students of your class**  
**P(x): x is perfect**  
**F(x): x is your friend**
- ① No one is perfect  
 ② All of your friends are perfect  
 ③ At least one of your friends is perfect.  
 ④ Not everyone is perfect
- b) Define Contraposition. Show that  $\neg(p \vee \neg(p \wedge q))$  is a Contradiction by using truth table. 1+4
- c) Using laws of logic, show that  $\neg(\neg p \wedge q) \wedge (p \vee q) \equiv p$  4
2. a) Consider Universal set U and two sets A and B. Shade the following sets using Venn diagram 2+3
- I.  $A \cap B'$   
 II.  $A' \cup B'$
- Prove that,  $(A \cup B)' = A' \cap B'$
- b) Find the Domain and Range of the following function:  $f(x) = \sqrt{4 - x^2}$  4
- c) The  $n^{\text{th}}$  term of a sequence is given by  $a_n = 3n + 2$  2+3
- ① Write the first three terms of the sequence.  
 ② Find the sum of the first 20 terms of the sequence.
3. a) State the Pigeonhole principle. Find the minimum number of students in a class to be sure that three of them are born in the same month. 1+2
- b) How many ways can the word MISSISSIPPI be arranged? 2.5
- c) A club has 15 members. How many ways can a president, a vice president, and a secretary be chosen? 2.5
- d) If two dice are rolled, what is the probability of getting a sum of 7? 3
- e) Find the coefficient of  $x^4$  in the expansion of  $(1+x)^7$ . 3
4. a) At least how many nodes are needed for making a graph? Draw a graph from adjacency matrix: 1+4
- $$A = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$$
- b) What is path? How many edges are there in a graph with 8 vertices each of degree 5? 1+3
- c) What are the differences between simple graph and multi-graph? 2
- d) Draw two graphs that are isomorphic and describe briefly. 3



5. a) Define Tree and Binary Tree with examples. Differentiate between Trees and Graphs. 4  
 b) What is spanning tree? Construct the minimum spanning tree for given graph using Prim's algorithm: 1+4



- c) Find the solution to the recurrence relation  $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$  with the initial conditions  $a_0 = 2, a_1 = 5, a_2 = 15$ . 5

6. a) What is relation? Define reflexive, symmetric and transitive relation. 1+3  
 b) Consider the following five relations on the set  $A = \{1, 2, 3, 4\}$ : 5

$$B = \{(1,1), (1,2), (2,3), (1,3), (4,4)\}$$

$$C = \{(1,2), (2,3), (3,1)\}$$

$$D = \{(1,2), (2,1), (1,1), (2,2)\}$$

$$E = \{(1,2), (3,4), (3,3)\}$$

$$F = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$$

Determine the types of above relations.

- c) What is Zero-One matrix? Find the join and meet of matrix A and B : 1+4

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

7. a) What are the Floor and Ceiling Functions? Describe with examples. 2  
 b) Define Universal quantifier and Existential quantifier. translate the following statements into logical expressions: 2+2

① All lions are fierce.

② Some lions don't drink coffee.

where  $P(x)$ ,  $Q(x)$ , and  $R(x)$  denote "x is a lion", "x is fierce" and "x drinks coffee" respectively.

- c) What are argument and fallacy? Using Contraposition, prove that: "If n is an integer and  $3n+2$  is odd, then n is odd". 2+3

- d) Among 50 students in a class, 26 got an A in the first examination and 21 got an A in the second examination. If 17 students did not get an A in either examination, how many students got an A in both the examination? 3